IEOR 4101: Probability, Statistics and Simulation

Lecture 3: Conditional Probability

Conditional Probability

- **Definition**: For two events A and B (with $P(B)>0$), the conditional probability $P(A|B)$ is defined to be

$$P(A|B) = P(A \cap B)/P(B)$$

- This is the probability of A given the knowledge that B occurs

Conditional Probability

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards?

$$\frac{\#\text{ event}}{\#\text{ sample space}}$$

$$= \frac{{}_4P_2}{{}_{52}P_2} \quad \text{or} \quad = \frac{\binom{4}{2}}{\binom{52}{2}}$$

$$= \frac{4 \times 3}{52 \times 51} \qquad = \frac{4 \times 3}{52 \times 51}$$

Conditional Probability

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards?
  - P(both being ace) = $\frac{{}_4P_2}{{}_{52}P_2} = \frac{4\times3}{52\times51}$

Conditional Probability

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards?
  - P(both being ace) = $\frac{{}_4P_2}{{}_{52}P_2} = \frac{4\times3}{52\times51}$

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards, given the first one is an ace?

A = 2 aces

B = 1st one is ace

$$P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{P(\text{2 aces})}{P(\text{1st one is ace})}$$

$$= \frac{\frac{4\times3}{52\times51}}{\frac{4}{52}} = \frac{3}{51}$$

Conditional Probability

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards?
  - P(both being ace) = $\frac{{}_4P_2}{{}_{52}P_2} = \frac{4\times 3}{52\times 51}$

- Ex. What's the probability of drawing 2 aces in a deck of 52 cards, given the first one is an ace?
  - P(first one being ace) = $\frac{4}{52}$
  - P(both being ace) = $\frac{{}_4P_2}{{}_{52}P_2} = \frac{4\times 3}{52\times 51}$
  - P(second one being ace | first one being ace) = $\frac{3}{51}$

Conditional Probability

- **Example**. A bin contains 5 defective, 10 partially defective and 25 acceptable transistors. A transistor is picked at random and put to use. If it does not immediately fail, what is the probability that it is acceptable?

$$P(\text{acceptable} \mid \text{does not immediately fail})$$

$$= \frac{P(\text{acceptable and does not immediately fail})}{P(\text{does not immediately fail})}$$

$$= \frac{P(\text{acceptable})}{P(\text{does not immediately fail})}$$

$$= \frac{25/40}{35/40} = \frac{25}{35}$$

Conditional Probability

- **Example**. A bin contains 5 defective, 10 partially defective and 25 acceptable transistors. A transistor is picked at random and put to use. If it does not immediately fail, what is the probability that it is acceptable?

- **Solution**. P(acceptable|not defective)
  = P(acceptable and not defective)/P(not defective)
  = P(acceptable)/P(not defective)
  = (25/40)/(35/40)
  = 5/7

Conditional Probability

- **Example**. In digital camera market, 60% buyers include optional memory card, 40% include extra battery, and 30% include both. Given a selected buyer purchased an extra battery, what is the probability that she also purchased an optional memory card?

$$P(\text{memory card} \mid \text{battery})$$

$$= \frac{P(\text{memory card and battery})}{P(\text{battery})}$$

$$= \frac{30\%}{40\%}$$

$$= \frac{3}{4}$$

Conditional Probability

- **Example**. In digital camera market, 60% buyers include optional memory card, 40% include extra battery, and 30% include both. Given a selected buyer purchased an extra battery, what is the probability that she also purchased an optional memory card?

- **Solution**. P(optional memory card | extra battery)
  = P(optional memory card and extra battery) / P(extra battery)
  = 30%/40%
  = 75%

Conditional Probability

- **Example**. In digital camera market, 60% buyers include optional memory card, 40% include extra battery, and 30% include both. Given a selected buyer purchased an extra battery, what is the probability that she also purchased an optional memory card?

- **Solution**. P(optional memory card | extra battery)
= P(optional memory card and extra battery) / P(extra battery)
= 30%/40%
= 75%

- What about P(extra battery | optional memory card)?

$$\frac{30\%}{60\%} = \frac{1}{2}$$

Conditional Probability

- **Example**. A family has two children. Given that there is at least one girl in the family, what's the probability that there are two girls?

$$P(2G \mid \text{at least } 1G)$$

$$= \frac{P(2G \text{ and at least } 1G)}{P(\text{at least } 1G)}$$

$$= \frac{P(2G)}{P(\text{at least } 1G)} \qquad \{GG, GB, BG, BB\}$$

$$= \frac{1/4}{3/4} = \frac{1}{3}$$

Conditional Probability

- **Example**. A family has two children. Given that there is at least one girl in the family, what's the probability that there are two girls?

- **Example**. A family has two children. Given that the older child is a girl, what's the probability that there are two girls?

$$P(2G \mid \text{older is } G)$$

$$= \frac{P(2G)}{P(\text{older is } G)}$$

$$= \frac{1/4}{2/4} = \frac{1}{2}$$

Conditional Probability

- Can rewrite the conditional probability formula as

$$P(A \cap B) = P(A|B)P(B)$$

- Sometimes called the **multiplication rule**
- Can expand the formula: $P(A \cap B \cap C) = P(C|A \cap B)P(B|A)P(A)$.

Conditional Probability

Example: You know that with 30% probability that your company will open a new branch in New York. You are also 60% certain that if the branch is opened you will become its new manager. What is the probability that you will be a new branch manager?

$30\% \times 60\%$

Conditional Probability

Example. You travel from New York to Singapore, with connecting flights in Chicago and Japan. You have one checked-in luggage that is transferred from one airplane to another. In New York, there is a 5% chance that your luggage is not placed in the right airplane. The probability is 3% in Chicago and 2% in Japan. What is the probability that your luggage does not reach Singapore with you? Given it does not reach Singapore, what is the probability that it was lost in Chicago?

P(not reach Singapore)

= P(lost in NY or lost in Chicago or lost in Japan)

= P(lost in NY) + P(lost in Chicago) + P(lost in Japan)

by mutual exclusiveness

$$P(\text{lost in NY}) = 5\%$$

$$P(\text{lost in Chicago})$$

$$= P(\text{pass NY})\, P(\text{lost in Chicago} \mid \text{pass NY})$$

$$= (1 - 5\%) \times 3\%$$

$$P(\text{lost in Japan})$$

$$= P(\text{pass NY})\, P(\text{pass Chicago} \mid \text{pass NY})\, P(\text{lost in Japan} \mid \text{pass NY and Chicago})$$

$$= (1-5\%) \times (1-3\%) \times 2\%$$

$P(\text{not reach Singapore})$

$= 5\% + (1-5\%)\times 3\% + (1-5\%)\times(1-3\%)\times 2\%$

Alternative approach:

$P(\text{not reach Singapore})$

$= 1 - P(\text{reach Singapore})$

$= 1 - (1-5\%)\times(1-3\%)\times(1-2\%)$

$$P(\text{lost in Chicago} \mid \text{not reach Singapore})$$

$$= \frac{P(\text{lost in Chicago})}{P(\text{not reach Singapore})}$$

$$= \frac{(1-5\%)\times 3\%}{5\% + (1-5\%)\times 3\% + (1-5\%)\times(1-3\%)\times 2\%}$$

Independence

- **Definition**: two events A and B are independent if

$$P(A \cap B) = P(A)P(B)$$

- Recall the conditional probability formula

$$P(A \cap B) = P(A|B)P(B)$$

If $P(A) = P(A|B)$, i.e., occurrence of A does not depend on that of B, then A is independent of B

- Thus, an equivalent definition of independence between A and B is $P(A)=P(A|B)$, and also $P(B)=P(B|A)$

- Note: if $P(A) = P(A|B)$, then $P(A) = P(A|B^c)$ (can you prove it?)

If A & B are indept, i.e. $P(A \cap B) = P(A)P(B)$

then

$$P(A \mid B^c) = \frac{P(A \cap B^c)}{P(B^c)}$$

$P(A) = P(A \cap B) +$
$P(A \cap B^c)$

$$= \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

because

$(A \cap B) \cup (A \cap B^c) = A$

and $(A \cap B)$ & $(A \cap B^c)$ are disjoint

$$= \frac{P(A) - P(A)P(B)}{1 - P(B)} = P(A)$$

The following are all equivalent definitions of independence:

$$P(A \cap B) = P(A)\,P(B)$$

$$P(A \mid B) = P(A)$$

$$P(B \mid A) = P(B)$$

$$P(A \mid B^c) = P(A)$$

$$P(B \mid A^c) = P(B)$$

$$P(B^c \mid A^c) = P(B^c) \quad \ldots$$

Independence

- **Example**: two fair dice are thrown. Let A be the event that the number shown by the first die is even, and B the event that the sum of the dice is odd. Are A and B independent?

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P(A \cap B) = P(1^{st} = \text{even},\ 2^{nd} = \text{odd})$$

$$= \frac{1}{4}$$

A and B are indept.

Independence

- **Example**: two fair dice are thrown. Let A be the event that the number shown by the first die is even, and B the event that the sum of the dice is odd. Are A and B independent?

- **Answer**: they are independent

- P(A) = 1/2
  P(B) = 1/2 – (odd, even) or (even, odd) both result in an odd sum; each happens with probability 9/36 = ¼
  P(A∩B) = P(first die is even and second die is odd) = 1/2*1/2 =1/4

Independence

- What about independence of more than 2 events?

- For three events A, B and C: they are independent if $P(A \cap B \cap C) = P(A)P(B)P(C)$, $P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$ and $P(A \cap C) = P(A)P(C)$

- For $A_1, A_2, \ldots, A_n$, they are independent if for every subcollection $A_{1'}, A_{2'}, \ldots, A_{r'}$, we have

$$P(A_{1'} \cap A_{2'} \cap \ldots \cap A_{r'}) = P(A_{1'})P(A_{2'})\ldots P(A_{r'})$$

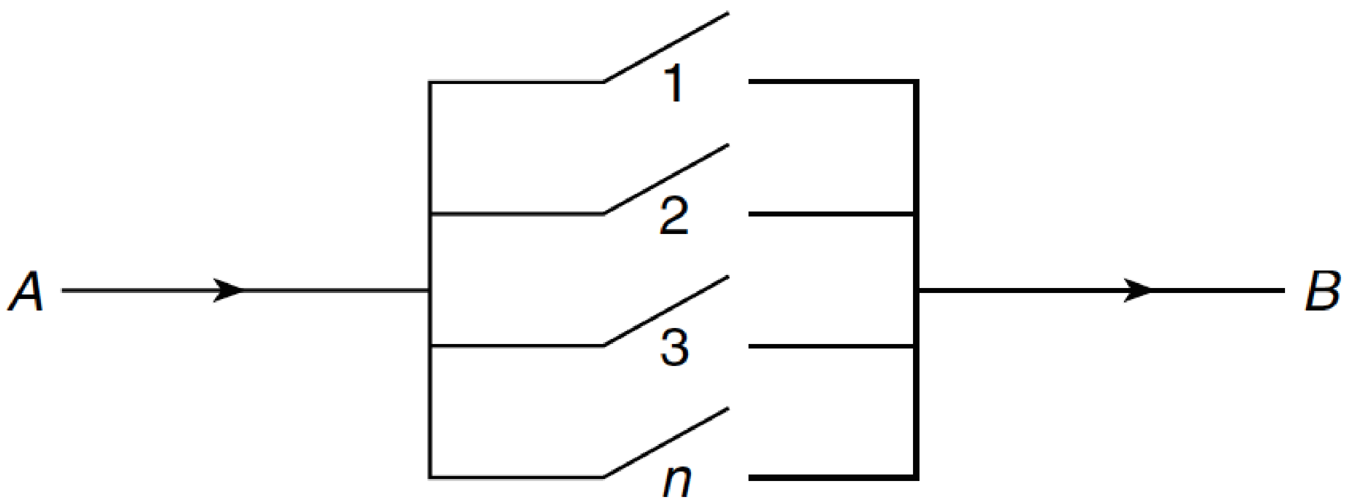
pairwise independence: For $A_1, A_2, \cdots, A_n$,

$$P(A_i \cap A_j) = P(A_i)P(A_j) \text{ for any } i, j$$

pairwise independence $\neq$ independence

Independence

- **Example**. A system with n components is called parallel if it functions when at least one component functions. For a parallel system, if each component functions with probability 90% independently, what is the probability that the system functions?



P(system function) = P(at least 1 component works) = 1 − P(no component works)

$$= 1 - P\left(\begin{array}{c}\text{component } 1\\ \text{fails}\end{array}\right) \times P\left(\begin{array}{c}\text{component } 2\\ \text{fails}\end{array}\right)$$

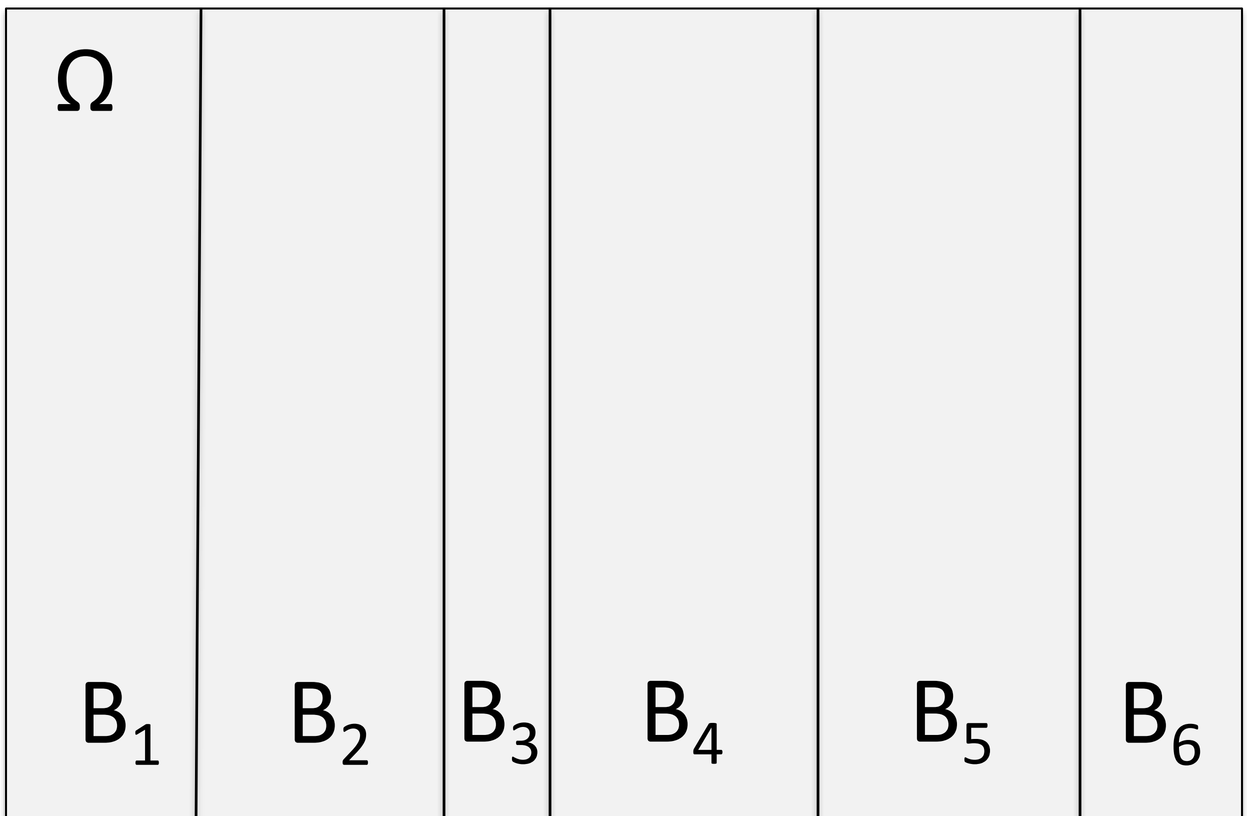
$$\times \ldots \times P\left(\begin{array}{c}\text{component } n\\ \text{fails}\end{array}\right)$$

by independance

$$= 1 - (0.1)^n$$

Law of Total Probability

- Let $B_1, B_2, \ldots, B_n$ be a collection of mutually exclusive events, so that whenever $i \neq j$, $B_i \cap B_j = \emptyset$
- Furthermore, suppose that $B_1 \cup B_2 \cup \ldots \cup B_n = \Omega$ is the whole sample space
- $B_1, B_2, \ldots, B_n$ said to be a *partition* of Ω



Law of Total Probability

- Let $B_1, B_2, \ldots, B_n$ be a collection of mutually exclusive events, so that whenever $i \neq j$, $B_i \cap B_j = \emptyset$
- Furthermore, suppose that $B_1 \cup B_2 \cup \ldots \cup B_n = \Omega$ is the whole sample space
- $B_1, B_2, \ldots, B_n$ said to be a *partition* of Ω
- Then for any event A,

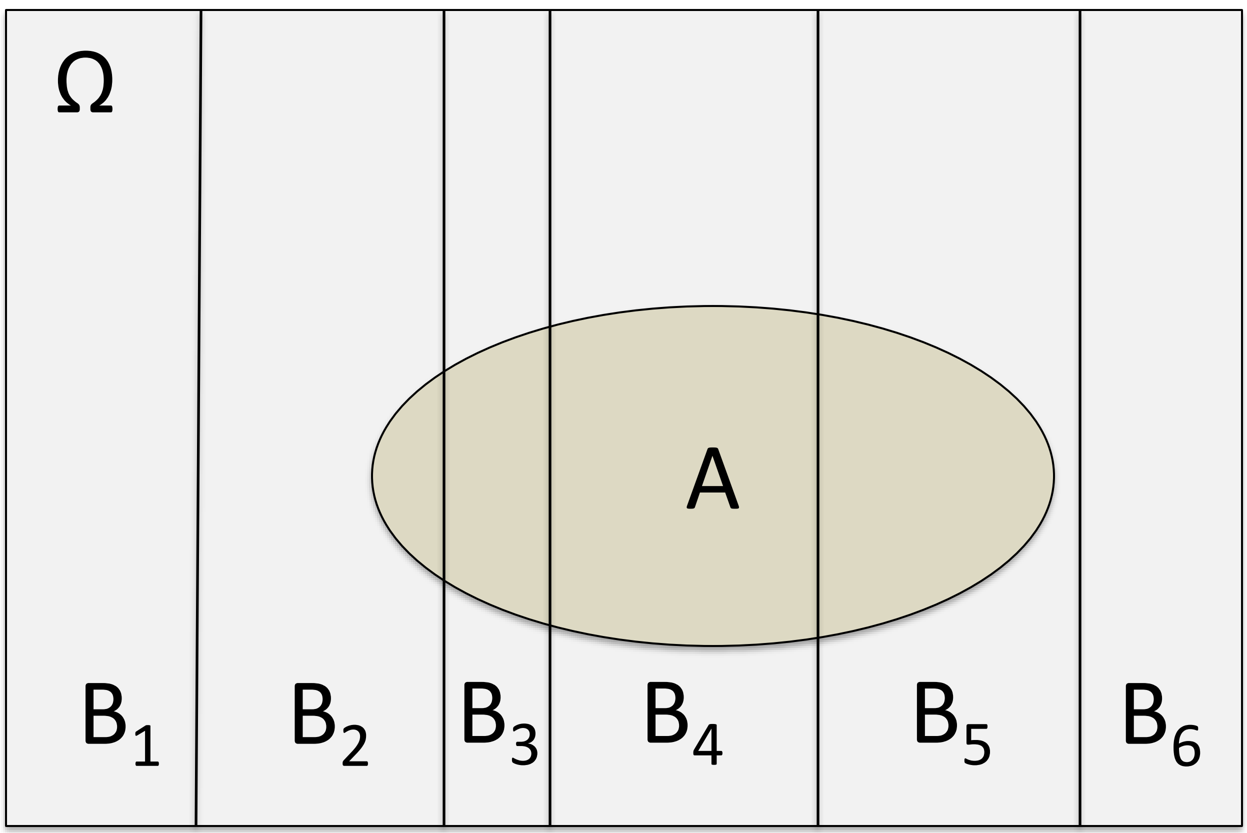
$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots + P(A \cap B_n);$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \ldots + P(A|B_n)P(B_n)$$

- Can extend to infinitely many $B_1, B_2, \ldots$

Law of Total Probability

- Let $B_1, B_2, \ldots, B_n$ be a collection of mutually exclusive events, so that whenever $i \neq j$, $B_i \cap B_j = \emptyset$
- Furthermore, suppose that $B_1 \cup B_2 \cup \ldots \cup B_n = \Omega$ is the whole sample space
- $B_1, B_2, \ldots, B_n$ said to be a *partition* of Ω

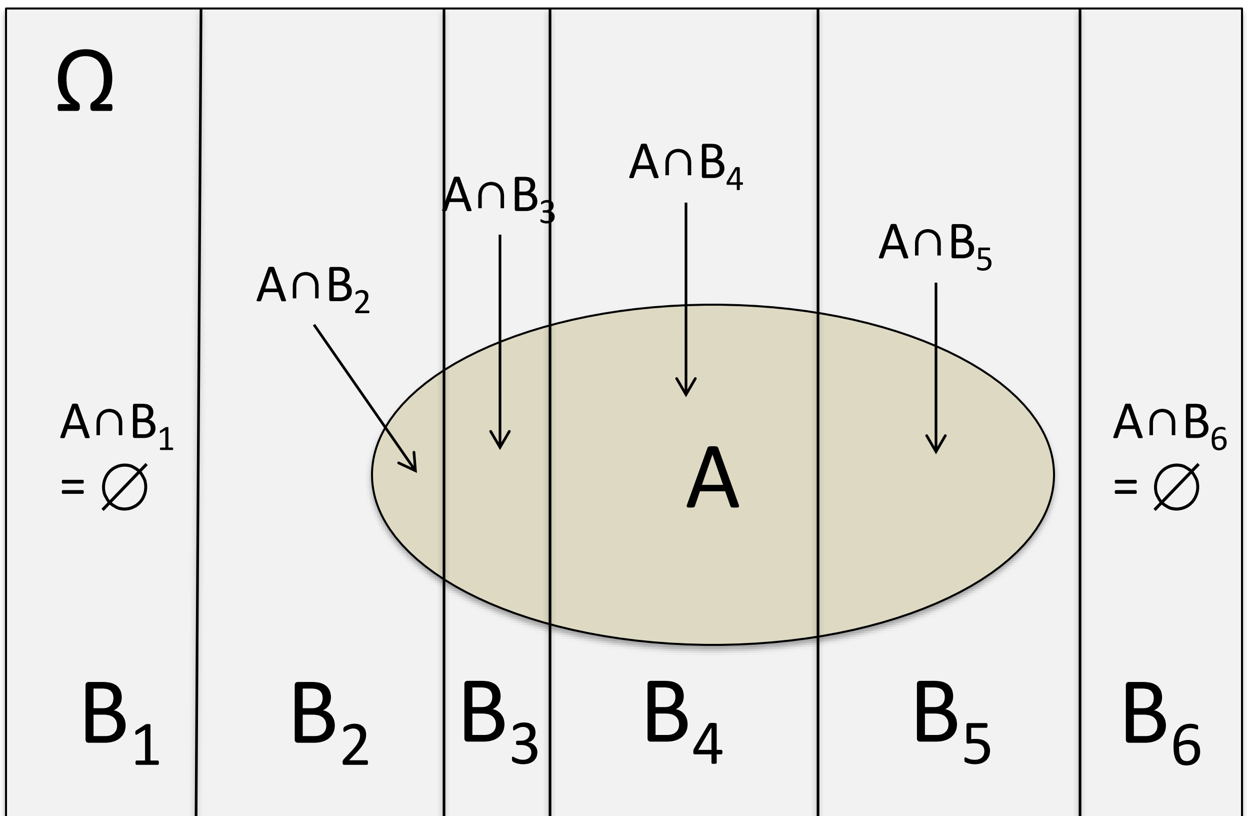


Law of Total Probability

- For A:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n);$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

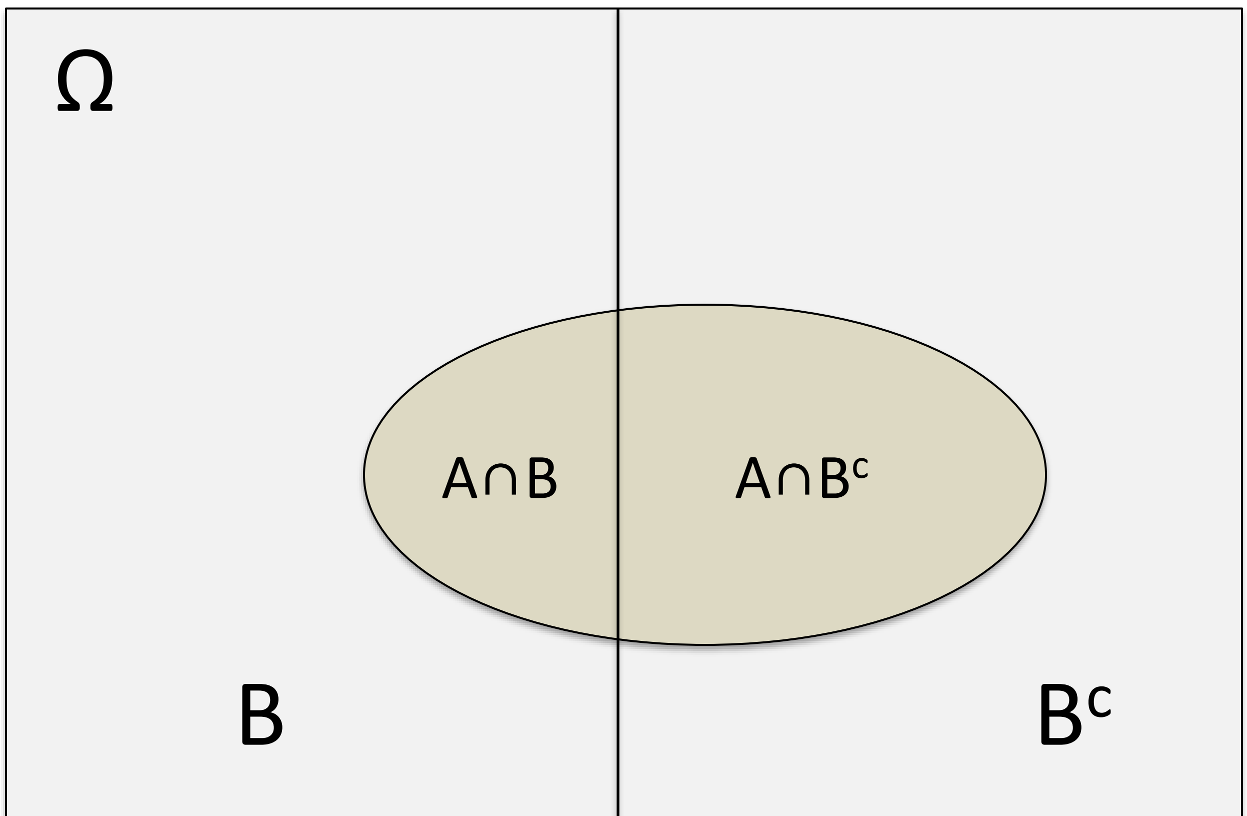


Law of Total Probability

- Special case:

$$P(A) = P(A \cap B) + P(A \cap B^c);$$

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$



Law of Total Probability

- **Example**. A die is rolled repeatedly to yield the number 1, and then a coin is tossed the same number of times. What is the probability that heads will not appear?

$A =$ no H

$B_1 =$ "1" appears first at 1^{st} roll

$B_2 =$ "1" appears first at 2^{nd} roll

$\vdots$

$B_k =$ "1" appears first at k^{th} roll

$\vdots$

$B_1, B_2, \ldots$ is a partition of Ω.

$$P(A) = P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) + \cdots$$

$$P(B_1) = \frac{1}{6} \qquad P(B_k) = \left(\frac{5}{6}\right)^{k-1} \times \frac{1}{6}$$

$$P(B_2) = \frac{5}{6} \times \frac{1}{6} \qquad \vdots$$

$$P(B_3) = \left(\frac{5}{6}\right)^2 \times \frac{1}{6}$$

$$\vdots$$

$$P(A \mid B_1) = \frac{1}{2}$$

$$P(A \mid B_2) = \left(\frac{1}{2}\right)^2$$

$$\vdots$$

$$P(A \mid B_k) = \left(\frac{1}{2}\right)^k$$

$$\vdots$$

$$P(A)$$

$$= \frac{1}{6} \times \frac{1}{2} +$$

$$\frac{5}{6} \times \frac{1}{6} \times \left(\frac{1}{2}\right)^2 +$$

$$\left(\frac{5}{6}\right)^2 \times \frac{1}{6} \times \left(\frac{1}{2}\right)^3 +$$

$$\vdots$$

$$\left(\frac{5}{6}\right)^{k-1} \times \frac{1}{6} \times \left(\frac{1}{2}\right)^k + \cdots$$

Law of Total Probability

- **Example**. A die is rolled repeatedly to yield the number 1, and then a coin is tossed the same number of times. What is the probability that heads will not appear?

- **Solution**. Let A be the event that heads will not appear, and B_i the event that the coin/die is tossed/rolled i times. Then
  $P(B_i) = (5/6)^{i-1}(1/6)$, i=1,2,3,…

  $P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + ...$
  $P(A|B_i) = (1/2)^i$
  $P(A) = (1/2)*(1/6) + (1/2)^2(5/6)(1/6) +…+ (5/12)^{i-1}(1/12) +…$
  $= (1/12)[1/(1-5/12)] = 1/7$